

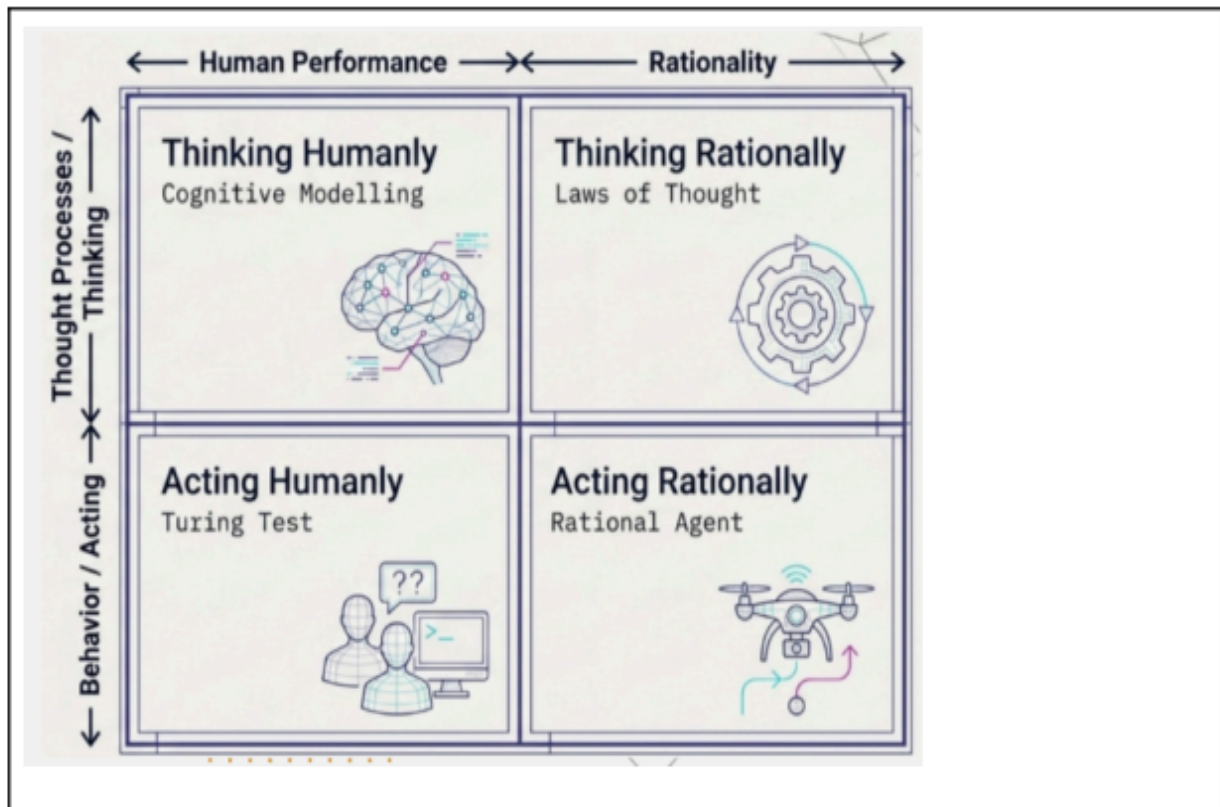
Question 1

- a) Artificial intelligence (A.I.) is a branch of computer science that focuses on building intelligent machines that function and respond just like people. There are four approaches to A.I. where they vary along two dimensions that are concerned with 1) thought vs. behaviour, and 2) human performance vs. rationality. For instance, one of them is the Turing test approach that intends to develop A.I. systems that are capable of acting like a human.

(i) Name the other **THREE (3)** approaches and briefly describe each of them. (3 + 3 marks)

- The cognitive modelling approach that intends to develop A.I. systems that are capable of thinking like human .
- The laws of thought approach intends to develop A.I. systems that are capable of thinking rationally .
- The rational agent approach intends to develop A.I. systems that are capable of acting rationally .

(ii) Draw a table with the use of labels to illustrate how these **FOUR (4)** approaches differ from each other based on the two dimensions. (6 marks)



- b) *ChatGPT* is a revolutionary new A.I. chat-bot that creates human-like conversational dialogue. Its achievement has surpassed many users' expectations and it already has more than 100 million active users two months after its first launch at the end of 2022. The *Turing test* is a test of a machine's (i.e., especially chat-bots) ability to exhibit intelligent behaviour equivalent to, or indistinguishable from, that of a human. There are many variants of this test and the *total Turing test* is one of them.

- (i) Identify how the *total Turing test* is different from the *Turing test*. (2 marks)

The total Turing test has two additional requirements added to create a stronger benchmark of intelligence:

- Perceptual ability (OR to derive information from its percepts)
- Object manipulation ability (OR to manipulate objects and move about)

- (ii) Identify whether *ChatGPT* can pass the *Turing test*. Justify your answer by focusing on the capabilities of *ChatGPT*. (1 + 4 marks)

Yes. The reason is that ChatGPT is capable of the following:

- Natural language processing (OR it can communicate in human languages)
- Knowledge representation (OR it can store what it knows or hears)
- Automated reasoning (OR it can do reasoning and draw new conclusions)
- Machine learning (OR it can adapt to new circumstances and handle uncertainties)

(iii) Identify whether ChatGPT can pass the *total Turing test*. Justify your answer by focusing on the capabilities of *ChatGPT*. (1 + 2 marks)

No. . The reason is that ChatGPT is not capable of either one or both If the following:

- Computer vision (OR it can derive information from its percepts)
- Robotics (OR it can manipulate objects and move about)

(iv) Which one is more relevant to the definition of A.I. provided in **Question 1 a)**, the *Turing test* or the *total Turing test*? Justify your answer. (1 + 2 marks)

The total Turing test. The reason is that human can derive information through vision and manipulating objects and move about

Question 2

- a) The world's first self-driving taxi service has been launched in Singapore in 2016. The *PEAS* description can be used to describe the problems to which the self-driving taxi solves.

- (i) Identify what the acronym *PEAS* stands for. (4 marks)

P: Performance measure

E: Environment

A: Actuators

S: Sensors

- (ii) Construct a *PEAS* description for this problem by providing **ONE (1)** example for each of the four elements. (4 marks)

Performance measure: Safe trip, fast trip, legal trip, comfortable trip, maximize profits, minimise impact on other road users

Environment: Roads, other traffic, police, pedestrians, customers, weather

Actuators: Steering, accelerator, brake, signal, horn, display, speech

Sensors: Cameras, radar, speedometer, GPS, engine sensors, accelerometer, microphones, touchscreen

- (iii) Identify **SEVEN (7)** properties of the environment. Justify your answer. (7 + 7 marks)

Environment	Justification
Fully observable	Due to the limitation of current image processing/ computer vision technology, the agent does not have access to the complete state of the road.

Multi-agent	There are many other drivers/vehicles on the road.
Non-deterministic	The agent has no control over the outcome of its action because the outcome is also influenced by the actions of the other agents.
Sequential	The choice of an action at any time has effect on the next action and the outcome.
Dynamic	The environment keeps changing when the agent is deliberating.
Continuous	There is an infinite number of states and movements (or choice of actions).
Known	The agent is aware of the traffic rules.

- (iv) Identify whether the self-driving taxi is a rational agent or an omniscient agent. Justify your answer. (1 + 2 marks)

A rational agent . The reason is that:

- Doesn't know the actual outcome of its actions
- Acts under uncertainty most of the time.
- Aims to maximize its performance by achieving the best expected outcome

Question 3

- a) In A.I., search techniques are universal problem-solving methods. Rational agents or problem-solving agents in A.I. mostly used these search strategies or algorithms to solve a specific problem and provide the best result. *Greedy-Best First (GBF)* and *Steepest-Ascent Hill Climbing (SAHC)* are two examples of popular search techniques.

(i) Briefly describe *GBF* and *SAHC*. (2 marks)

- GBF is an informed search strategy that expands any node that is closest to the goal .
- SAHC is a local search strategy that keeps moving to the neighboring state with the best heuristic value

(ii) Identify **TWO (2)** commonalities between *GBF* and *SAHC*. (2 marks)

- Not cost-optimal
- Greedy strategy
- Soly rely on heuristic

(iii) Discuss **TWO (2)** differences between *GBF* and *SAHC*. (4 marks)

GBF	SAHC
Keeps track of expanded and unexpanded nodes	Keeps track of the current node (OR current state) only
Its solution is a sequence of actions.	Its solution is the goal state itself
Tree-like search is incomplete but graph search is complete	Incomplete

Aims to solve path searching problem	Aims to solve optimization problem
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- (iv) Suppose you are driving in a city. Identify whether *GBF* or *SAHC* is more appropriate to be deployed in solving the problem of searching for a route from one location to another. Justify your answer. (1 + 2 marks)

GBF, the reason is that the solution generated is a **sequence of actions** which can be used as a **guide to drive within the city**.